ĐẠI HỌC HOA SEN

TÔN TRỌNG SỰ KHÁC BIỆT

About
Data Analytics
Business Intelligence

Faculty of Information Technology

ĐẠI HỌC HOA SEN

Objectives

- ✓ Concept
- ✓ Roles in Organizations
- ✓ Characteristics
- ✓ Process
- ✓ Types of Analysis

Concepts

- **Data Analytics (DA)**
- **Data Analytics** is the process of inspecting, cleaning, transforming, and modeling data to discover useful insights, patterns, trends, and support decision-making.
  It focuses on **explaining, predicting, and optimizing** outcomes using statistical and computational techniques.
- Answers: *"What happened, why did it happen, what will happen, and what should we do?"*

- **Business Intelligence (BI)**
- **Business Intelligence** refers to technologies, architectures, and practices used to collect, integrate, analyze, and present business information to support operational and strategic decisions.
- BI emphasizes **reporting, dashboards, monitoring KPIs**, and structured decision support.
- Answers: *"What is happening in the business right now?*

Role in Organizations

- **Data Analytics**
  - Supports **strategic decision-making**
  - Identifies hidden patterns and trends
  - Enables forecasting and optimization
  - Supports AI/ML initiatives
  - Drives data-driven innovation
- Used by:
  - Data Analysts
  - Data Scientists
  - Strategy teams
  - Product teams

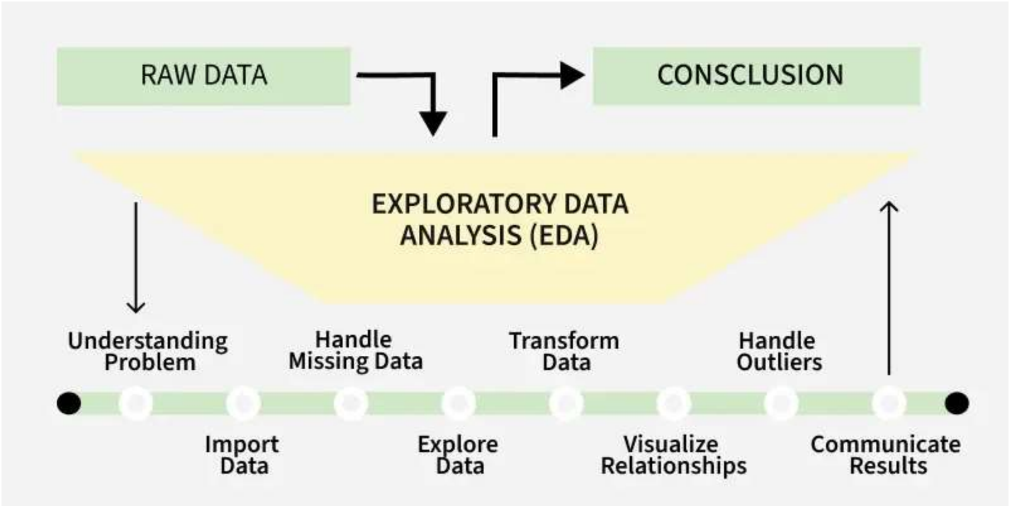
- **Business Intelligence**
  - Monitors performance through KPIs
  - Supports operational management
  - Enables data-driven daily decisions
  - Standardizes reporting processes
  - Improves transparency across departments
- Used by:
  - Executives
  - Managers
  - Business users
  - Operations teams

Characteristics

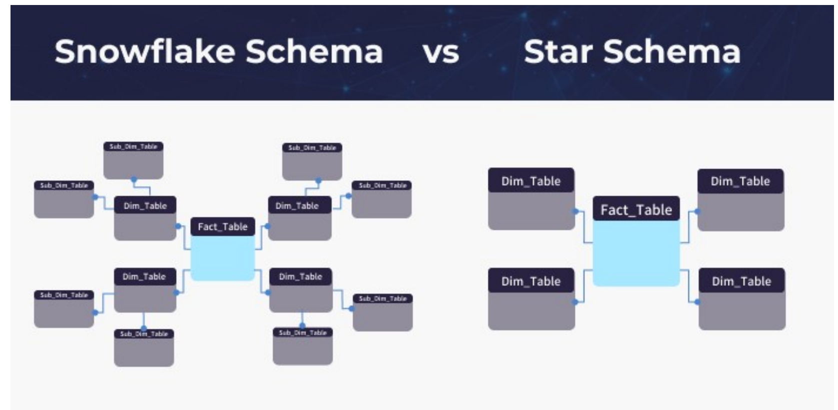
Aspect	Data Analytics	Business Intelligence
Focus	Insight & prediction	Reporting & monitoring
Data	Structured & unstructured	Mostly structured
Time orientation	Past + Future	Mainly past & present
Techniques	Statistical models, ML	Aggregation, OLAP
Output	Insights, models, forecasts	Reports, dashboards
Complexity	Higher	Moderate

Process

- **Data Analytics**
  - Define business problem
  - Collect data
  - Clean & preprocess data
  - Explore data (EDA - exploratory data analysis)
  - Apply statistical / ML models
  - Interpret & communicate results
  - Deploy model (if needed)
- Often follows frameworks like:
  - CRISP-DM (Cross Industry Standard Process for Data Mining)

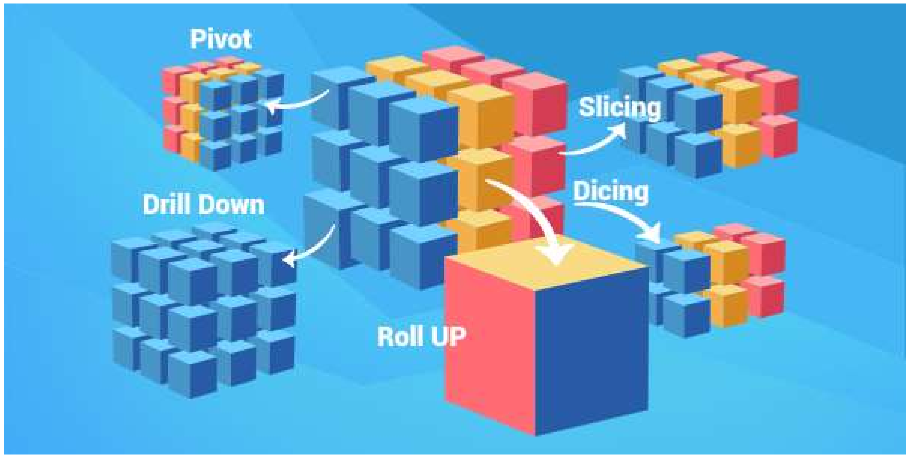


- **Business Intelligence**
  - Data collection from operational systems
  - ETL (Extract – Transform – Load)
  - Store in Data Warehouse
  - Data modeling (star/snowflake schema)
  - Reporting & dashboard creation
  - Performance monitoring (KPIs)



Types of Analysis

- **Data Analytics**
  - **Descriptive Analytics** – What happened? (Summarizes historical data)
  - **Diagnostic Analytics** – Why did it happen? (Explains causes)
  - **Predictive Analytics** – What will happen? (Forecasts future events)
  - **Prescriptive Analytics** – What should we do? (Recommends actions)



- **Business Intelligence**
  Primarily:
  - Descriptive analytics
  - Basic diagnostic analysis
  - KPI tracking
  - Trend analysis
  - Drill-down & slice-and-dice (OLAP)

  BI rarely includes advanced predictive modeling

Retail case study

- Company: Supermarket Chain
- BI: Sales dashboard by region
- DA: Demand forecasting model
- Outcome: Reduce stock-outs by 20%

Banking Case Study

- Company: Commercial Bank
- BI: Loan performance dashboard
- DA: Credit risk prediction model
- Outcome: Reduce default rate

E-commerce Case Study

- Company: Online Marketplace
- BI: Traffic & conversion dashboard
- DA: Recommendation system
- Outcome: Increase revenue per user

Tools

- **Data Analytics**
- Programming:
  - Python
  - R
  - SQL
- Libraries:
  - Pandas
  - NumPy
  - Scikit-learn
  - TensorFlow
- Platforms:
  - Jupyter Notebook
  - Apache Spark

- **Business Intelligence**
  - Microsoft Power BI
  - Tableau
  - Qlik Sense
  - SAP BusinessObjects

Strengths & Weaknesses

Criteria	Data Analytics	Business Intelligence
Strengths	Deep insights	Easy-to-use dashboards
	Predictive capability	Fast decision support
	Supports AI/ML	Standardized reporting
	Competitive advantage	User-friendly
Weaknesses	Requires technical skills	Limited predictive power
	More time-consuming	Less flexible for complex modeling
	Higher implementation cost	Mainly historical focus

Summary

Dimension	Data Analytics	Business Intelligence
Purpose	Discover & predict	Monitor & report
Users	Analysts, Data Scientists	Managers, Executives
Complexity	High	Medium
Time Focus	Past + Future	Past + Present
Output	Insights, Models	Dashboards, Reports

Data Analytics answers:

"Why, what next, and what should we do?"

BI answers:

"What is happening?"

- DA = Advanced analytical layer
- BI = Foundation

Exercise 1 – Classification Activity

- Students classify scenarios as DA or BI
- Example: Sales dashboard?
- Example: Predict churn probability?

Exercise 2 – Mini Case Analysis

- Retail company problem
- Identify BI components
- Identify DA opportunities
- Present group findings


Q&A